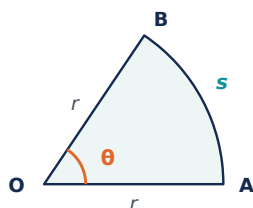


1 Measurement of Angles



arc $s = r\theta$

area $= \frac{1}{2}r^2\theta = \frac{1}{2}rs$

(θ must be in radians)

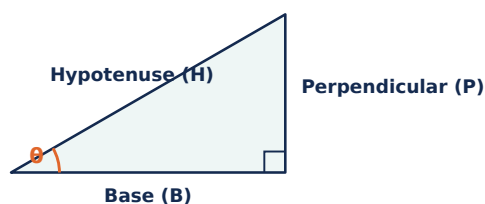
$\pi \text{ rad} = 180^\circ$

Radian measure — the angle subtended at the centre by an arc equal in length to the radius.

- 1 right angle $= 90^\circ = 100^g$ (grades) $= \frac{\pi}{2}$ radians.
- $\pi \text{ rad} = 180^\circ$; $1 \text{ rad} = \frac{180^\circ}{\pi} \approx 57^\circ 17' 45''$.
- Degrees D , grades G , radians θ : $\frac{D}{90} = \frac{G}{100} = \frac{2\theta}{\pi}$.
- Arc length $s = r\theta$; area of sector $= \frac{1}{2}r^2\theta = \frac{1}{2}rs$.

Non-negotiable: $s = r\theta$, the sector area, and every calculus formula ($\frac{d}{dx} \sin x = \cos x$, $\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$) are valid **only with θ in radians**. Convert first, always.

2 The Six Ratios & Their Reciprocals



$\sin \theta = P/H$

$\cos \theta = B/H$

$\tan \theta = P/B$

$\cot \theta = B/P$

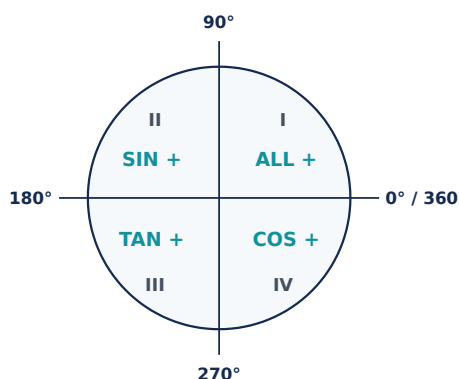
$\sec \theta = H/B$

$\text{cosec } \theta = H/P$

The six ratios in a right triangle. The three in teal are simply the reciprocals of the three above them.

- $\text{cosec } \theta = \frac{1}{\sin \theta}$, $\sec \theta = \frac{1}{\cos \theta}$, $\cot \theta = \frac{1}{\tan \theta}$.
- $\tan \theta = \frac{\sin \theta}{\cos \theta}$, $\cot \theta = \frac{\cos \theta}{\sin \theta}$.

3 Signs in the Four Quadrants



A - S - T - C

A → All positive (I)

S → sin, cosec (II)

T → tan, cot (III)

C → cos, sec (IV)

Everything else is negative in that quadrant.

Signs of the ratios — “Add Sugar To Coffee”. A ratio and its reciprocal always carry the same sign.

- A ratio and its reciprocal **always** share the same sign (e.g. if $\sin \theta < 0$ then $\operatorname{cosec} \theta < 0$).
- sin and cos are the “primary” signs; $\tan = \frac{\sin}{\cos}$ is positive exactly where sin and cos agree in sign (quadrants I and III).

4 Fundamental Identities

$$\sin^2 \theta + \cos^2 \theta = 1$$

$$1 + \tan^2 \theta = \sec^2 \theta$$

$$1 + \cot^2 \theta = \operatorname{cosec}^2 \theta$$

- Useful rearrangements: $\sec^2 \theta - \tan^2 \theta = 1$ and $\operatorname{cosec}^2 \theta - \cot^2 \theta = 1$.
- Factorised forms (used constantly): $(\sec \theta - \tan \theta)(\sec \theta + \tan \theta) = 1$; $(\operatorname{cosec} \theta - \cot \theta)(\operatorname{cosec} \theta + \cot \theta) = 1$.
- $\sin^4 \theta + \cos^4 \theta = 1 - 2 \sin^2 \theta \cos^2 \theta$; $\sin^6 \theta + \cos^6 \theta = 1 - 3 \sin^2 \theta \cos^2 \theta$.

5 Standard Values

θ	0°	30°	45°	60°	90°
$\sin \theta$	0	$\frac{1}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{\sqrt{3}}{2}$	1
$\cos \theta$	1	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{1}{2}$	0
$\tan \theta$	0	$\frac{1}{\sqrt{3}}$	1	$\sqrt{3}$	∞

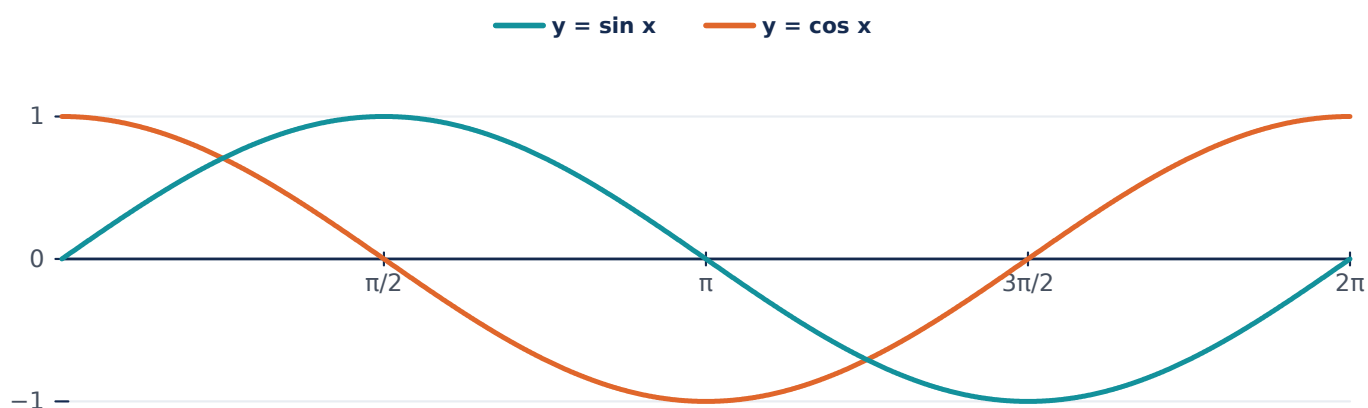
Memory aid: write $\sin \theta = \sqrt{\frac{n}{4}}$ for $n = 0, 1, 2, 3, 4$ across the row — that is 0, $\frac{1}{2}$, $\frac{1}{\sqrt{2}}$, $\frac{\sqrt{3}}{2}$, 1. Read the same list backwards for cos.

6 Domain, Range & Period

FUNCTION	DOMAIN	RANGE	PERIOD
$\sin \theta, \cos \theta$	$\mathbb{R}$	$[-1, 1]$	2π
$\tan \theta$	$\mathbb{R} - \{(2n+1)\frac{\pi}{2}\}$	$\mathbb{R}$	π
$\cot \theta$	$\mathbb{R} - \{n\pi\}$	$\mathbb{R}$	π
$\sec \theta$	$\mathbb{R} - \{(2n+1)\frac{\pi}{2}\}$	$(-\infty, -1] \cup [1, \infty)$	2π
$\operatorname{cosec} \theta$	$\mathbb{R} - \{n\pi\}$	$(-\infty, -1] \cup [1, \infty)$	2π

Period of $f(a\theta + b) = \frac{T}{|a|}$. So $\sin 2\theta$ has period π , and $|\sin \theta|$ has period π (not 2π).

7 Graphs of Sine and Cosine



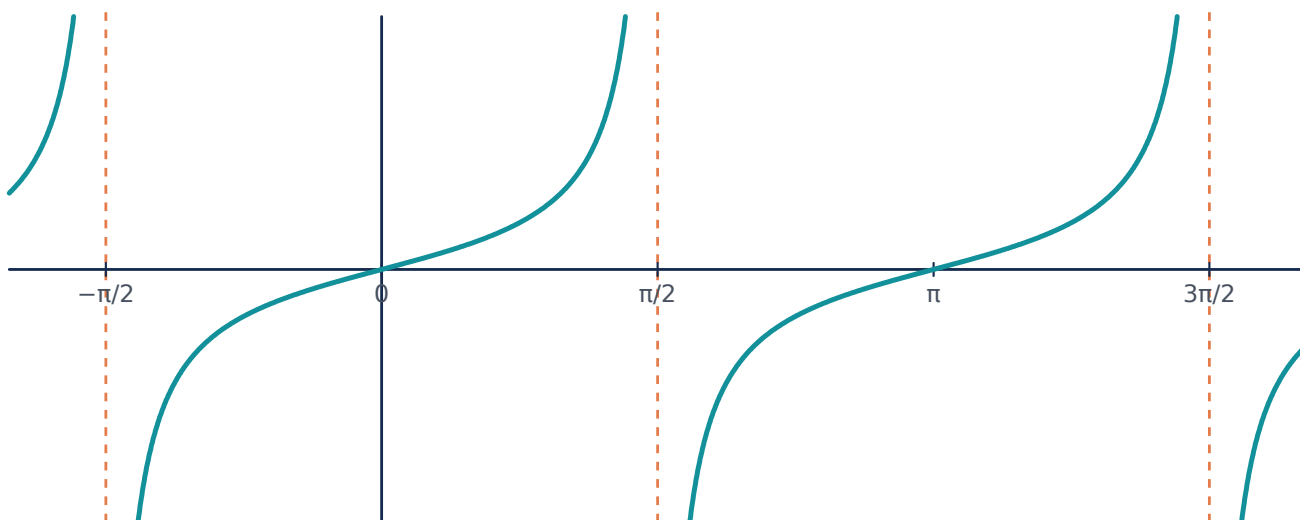
One full period. The cosine graph is the sine graph shifted left by $\pi/2$: $\cos x = \sin(x + \pi/2)$.

- $\sin \theta$ is **odd** (symmetric about the origin); $\cos \theta$ is **even** (symmetric about the y -axis).
- $\cos \theta = \sin(\theta + \frac{\pi}{2})$ — the same curve, shifted.
- Both are bounded: $-1 \leq \sin \theta, \cos \theta \leq 1$, which is the source of most maximum–minimum questions.

8 Graph of the Tangent

asymptotes at $x = (2n+1)\pi/2$

$y = \tan x$

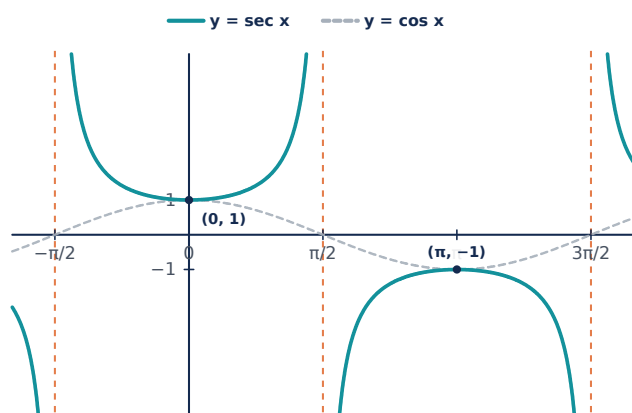


Period π (not 2π), range all of $\mathbb{R}$, and a vertical asymptote wherever $\cos x = 0$.

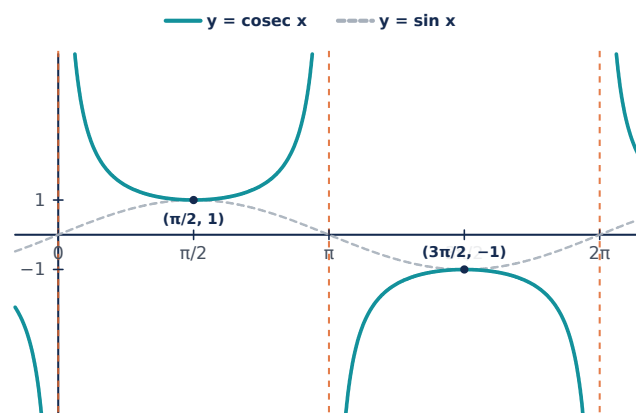
- $\tan \theta$ and $\cot \theta$ have period π ; the other four have period 2π .
- $\tan \theta$ is increasing on each branch, and takes **every** real value on each branch — which is exactly why $\tan \theta = \tan \alpha \Rightarrow \theta = n\pi + \alpha$.
- $\sec \theta$ and $\csc \theta$ never take values strictly between -1 and 1 .

Graphs of the Reciprocal Functions

Read each reciprocal against its faint partner. A reciprocal curve **shoots off to $\pm\infty$ exactly where its partner crosses 0**, and **touches ± 1 exactly where the partner peaks at ± 1** . That single idea reproduces all three shapes.



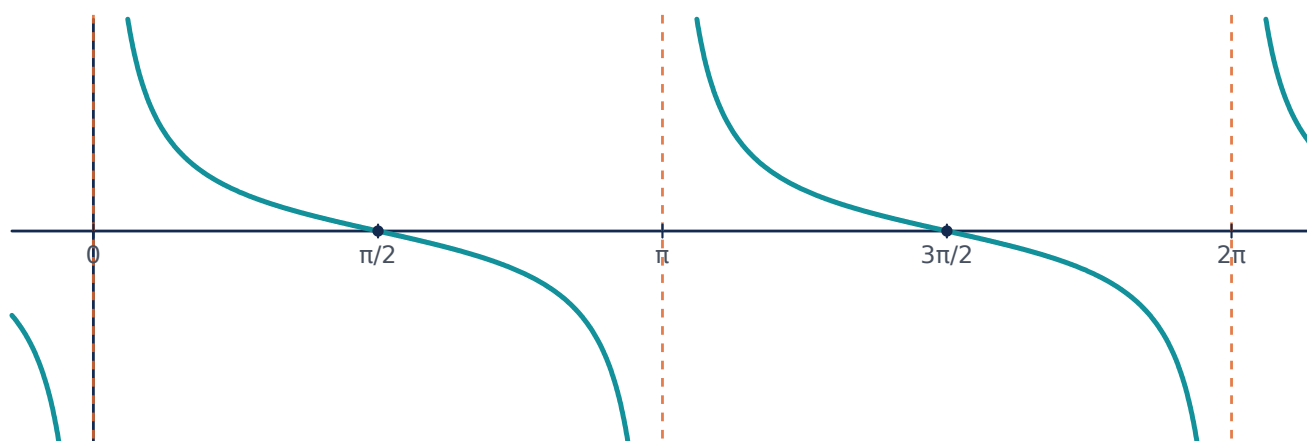
$y = \sec x = \frac{1}{\cos x}$. Each branch is a U (or n) whose vertex rests on $\cos x = \pm 1$; asymptotes where $\cos x = 0$.



$y = \operatorname{cosec} x = \frac{1}{\sin x}$. The same shape as secant, shifted by $\pi/2$; asymptotes where $\sin x = 0$.

$y = \cot x$

asymptotes at $x = n\pi$



$y = \cot x = \frac{\cos x}{\sin x}$. Strictly **decreasing** on every branch, zero at the odd multiples of $\pi/2$, asymptotes wherever $\sin x = 0$ — the mirror-image companion of the tangent.

FUNCTION	DOMAIN	RANGE	PERIOD	SYMMETRY
$\sec x$	$\mathbb{R} - \{(2n+1)\frac{\pi}{2}\}$	$(-\infty, -1] \cup [1, \infty)$	2π	even (like $\cos$)
$\operatorname{cosec} x$	$\mathbb{R} - \{n\pi\}$	$(-\infty, -1] \cup [1, \infty)$	2π	odd (like $\sin$)
$\cot x$	$\mathbb{R} - \{n\pi\}$	$\mathbb{R}$	π	odd (like $\tan$)

The one trap that costs marks: $\sec x$ and $\operatorname{cosec} x$ **never** take a value in $(-1, 1)$. Any equation forcing $\sec x = \frac{1}{2}$ or $\operatorname{cosec} x = 0$ has **no solution** — spot it before you start solving. Also useful: $\sec x = \operatorname{cosec}(x + \frac{\pi}{2})$, so the cosecant graph is just the secant graph slid right by $\pi/2$ — no need to memorise both.

9 Allied Angles — The Reduction Rule

To simplify $\sin(90^\circ \pm \theta)$, $\cos(180^\circ - \theta)$ and so on, use two steps.

- **Step 1 (does the name change?)** For $90^\circ \pm \theta$ and $270^\circ \pm \theta$ the ratio **changes**: $\sin \leftrightarrow \cos$, $\tan \leftrightarrow \cot$, $\sec \leftrightarrow \csc$. For $180^\circ \pm \theta$ and $360^\circ \pm \theta$ the ratio **stays the same**.
- **Step 2 (what sign?)** Treat θ as acute, find the quadrant of the whole angle, and take the sign the **original** ratio has there (ASTC).

ANGLE	SIN	COS	TAN
$-\theta$	$-\sin \theta$	$\cos \theta$	$-\tan \theta$
$90^\circ - \theta$	$\cos \theta$	$\sin \theta$	$\cot \theta$
$90^\circ + \theta$	$\cos \theta$	$-\sin \theta$	$-\cot \theta$
$180^\circ - \theta$	$\sin \theta$	$-\cos \theta$	$-\tan \theta$
$180^\circ + \theta$	$-\sin \theta$	$-\cos \theta$	$\tan \theta$
$270^\circ - \theta$	$-\cos \theta$	$-\sin \theta$	$\cot \theta$
$270^\circ + \theta$	$-\cos \theta$	$\sin \theta$	$-\cot \theta$
$360^\circ - \theta$	$-\sin \theta$	$\cos \theta$	$-\tan \theta$

10 Compound Angles

- $\sin(A \pm B) = \sin A \cos B \pm \cos A \sin B$
- $\cos(A \pm B) = \cos A \cos B \mp \sin A \sin B$
- $\tan(A \pm B) = \frac{\tan A \pm \tan B}{1 \mp \tan A \tan B}$ $\cot(A \pm B) = \frac{\cot A \cot B \mp 1}{\cot B \pm \cot A}$

Two results worth memorising

- $\sin(A + B) \sin(A - B) = \sin^2 A - \sin^2 B = \cos^2 B - \cos^2 A$
- $\cos(A + B) \cos(A - B) = \cos^2 A - \sin^2 B = \cos^2 B - \sin^2 A$

Three angles

$$\tan(A + B + C) = \frac{\tan A + \tan B + \tan C - \tan A \tan B \tan C}{1 - \tan A \tan B - \tan B \tan C - \tan C \tan A}$$

11 Multiple Angles — $2A$ and $3A$

Double angle

- $\sin 2A = 2 \sin A \cos A = \frac{2 \tan A}{1 + \tan^2 A}$
- $\cos 2A = \cos^2 A - \sin^2 A = 2 \cos^2 A - 1 = 1 - 2 \sin^2 A = \frac{1 - \tan^2 A}{1 + \tan^2 A}$
- $\tan 2A = \frac{2 \tan A}{1 - \tan^2 A}$

Triple angle

- $\sin 3A = 3 \sin A - 4 \sin^3 A$
- $\cos 3A = 4 \cos^3 A - 3 \cos A$
- $\tan 3A = \frac{3 \tan A - \tan^3 A}{1 - 3 \tan^2 A}$

Power-reduction (needed in every integration): $\sin^2 A = \frac{1 - \cos 2A}{2}$, $\cos^2 A = \frac{1 + \cos 2A}{2}$, $\sin^3 A = \frac{3 \sin A - \sin 3A}{4}$, $\cos^3 A = \frac{3 \cos A + \cos 3A}{4}$.

12 Sub-multiple (Half) Angles

- $\sin A = 2 \sin \frac{A}{2} \cos \frac{A}{2}$; $\cos A = 1 - 2 \sin^2 \frac{A}{2} = 2 \cos^2 \frac{A}{2} - 1$
- $\sin \frac{A}{2} = \pm \sqrt{\frac{1 - \cos A}{2}}$, $\cos \frac{A}{2} = \pm \sqrt{\frac{1 + \cos A}{2}}$
- $\tan \frac{A}{2} = \frac{\sin A}{1 + \cos A} = \frac{1 - \cos A}{\sin A} = \pm \sqrt{\frac{1 - \cos A}{1 + \cos A}}$

The t -substitution ($t = \tan \frac{A}{2}$)

$$\sin A = \frac{2t}{1 + t^2}$$

$$\cos A = \frac{1 - t^2}{1 + t^2}$$

$$\tan A = \frac{2t}{1 - t^2}$$

The $\pm$ is decided by the quadrant of $\frac{A}{2}$, **not** of A — a favourite JEE trap.

13 Sum ↔ Product Transformations

Sum / difference into product

- $\sin C + \sin D = 2 \sin \frac{C+D}{2} \cos \frac{C-D}{2}$
- $\sin C - \sin D = 2 \cos \frac{C+D}{2} \sin \frac{C-D}{2}$
- $\cos C + \cos D = 2 \cos \frac{C+D}{2} \cos \frac{C-D}{2}$
- $\cos C - \cos D = -2 \sin \frac{C+D}{2} \sin \frac{C-D}{2}$

Product into sum

- $2 \sin A \cos B = \sin(A+B) + \sin(A-B)$
- $2 \cos A \sin B = \sin(A+B) - \sin(A-B)$
- $2 \cos A \cos B = \cos(A+B) + \cos(A-B)$
- $2 \sin A \sin B = \cos(A-B) - \cos(A+B)$

Note the **minus sign** in $\cos C - \cos D$ — it is the one students drop most often.

14 Ratios of Special Angles

ANGLE	SIN	COS	TAN
15°	$\frac{\sqrt{3}-1}{2\sqrt{2}}$	$\frac{\sqrt{3}+1}{2\sqrt{2}}$	$2-\sqrt{3}$
18°	$\frac{\sqrt{5}-1}{4}$	$\frac{\sqrt{10+2\sqrt{5}}}{4}$	—
22½°	$\frac{\sqrt{2}-\sqrt{2}}{2}$	$\frac{\sqrt{2+\sqrt{2}}}{2}$	$\sqrt{2}-1$
36°	$\frac{\sqrt{10-2\sqrt{5}}}{4}$	$\frac{\sqrt{5}+1}{4}$	—
54°	$\frac{\sqrt{5}+1}{4}$	$\frac{\sqrt{10-2\sqrt{5}}}{4}$	—
75°	$\frac{\sqrt{3}+1}{2\sqrt{2}}$	$\frac{\sqrt{3}-1}{2\sqrt{2}}$	$2+\sqrt{3}$

Remember just two: $\sin 18^\circ = \cos 72^\circ = \frac{\sqrt{5}-1}{4}$ and $\cos 36^\circ = \sin 54^\circ = \frac{\sqrt{5}+1}{4}$. Everything else in the 18° family follows from $\sin^2 + \cos^2 = 1$.

15 Maximum & Minimum Values

Write $a \sin \theta + b \cos \theta = R \sin(\theta + \alpha)$ with $R = \sqrt{a^2 + b^2}$ and $\tan \alpha = \frac{b}{a}$. Then

$$-\sqrt{a^2 + b^2} \leq a \sin \theta + b \cos \theta \leq \sqrt{a^2 + b^2}$$

- $a \sin^2 \theta + b \cos^2 \theta$ lies between $\min(a, b)$ and $\max(a, b)$.
- $\sin \theta \cos \theta = \frac{1}{2} \sin 2\theta \Rightarrow$ maximum $\frac{1}{2}$, minimum $-\frac{1}{2}$.
- $\sec^2 \theta + \operatorname{cosec}^2 \theta \geq 4$; $\tan^2 \theta + \cot^2 \theta \geq 2$.
- For θ in a **restricted** interval, always check the endpoints as well as the turning point.

16 Conditional Identities ($A + B + C = \pi$)

These hold whenever A, B, C are the angles of a triangle.

- $\sin 2A + \sin 2B + \sin 2C = 4 \sin A \sin B \sin C$
- $\cos 2A + \cos 2B + \cos 2C = -1 - 4 \cos A \cos B \cos C$
- $\sin A + \sin B + \sin C = 4 \cos \frac{A}{2} \cos \frac{B}{2} \cos \frac{C}{2}$
- $\cos A + \cos B + \cos C = 1 + 4 \sin \frac{A}{2} \sin \frac{B}{2} \sin \frac{C}{2}$
- $\tan A + \tan B + \tan C = \tan A \tan B \tan C$
- $\cot A \cot B + \cot B \cot C + \cot C \cot A = 1$
- $\tan \frac{A}{2} \tan \frac{B}{2} + \tan \frac{B}{2} \tan \frac{C}{2} + \tan \frac{C}{2} \tan \frac{A}{2} = 1$
- $\cot \frac{A}{2} + \cot \frac{B}{2} + \cot \frac{C}{2} = \cot \frac{A}{2} \cot \frac{B}{2} \cot \frac{C}{2}$

17 Trigonometric Series (Angles in AP)

For n terms with first angle α and common difference β :

$$\sum_{k=0}^{n-1} \sin(\alpha + k\beta) = \frac{\sin \frac{n\beta}{2}}{\sin \frac{\beta}{2}} \cdot \sin \left(\alpha + \frac{(n-1)\beta}{2} \right)$$

$$\sum_{k=0}^{n-1} \cos(\alpha + k\beta) = \frac{\sin \frac{n\beta}{2}}{\sin \frac{\beta}{2}} \cdot \cos \left(\alpha + \frac{(n-1)\beta}{2} \right)$$

Identical except for the last factor. Both fail when $\sin \frac{\beta}{2} = 0$ (i.e. β a multiple of 2π) — then every term is equal and the sum is simply $n \sin \alpha$ or $n \cos \alpha$.

18 Standard Product Identities

- $\sin \theta \sin(60^\circ - \theta) \sin(60^\circ + \theta) = \frac{1}{4} \sin 3\theta$
- $\cos \theta \cos(60^\circ - \theta) \cos(60^\circ + \theta) = \frac{1}{4} \cos 3\theta$
- $\tan \theta \tan(60^\circ - \theta) \tan(60^\circ + \theta) = \tan 3\theta$

The doubling chain

$$\cos \theta \cdot \cos 2\theta \cdot \cos 4\theta \cdots \cos(2^{n-1}\theta) = \frac{\sin(2^n \theta)}{2^n \sin \theta}$$

Proof idea: multiply and divide by $2 \sin \theta$ and use $2 \sin \theta \cos \theta = \sin 2\theta$ repeatedly — the whole product telescopes.

19 Trigonometric Equations — General Solutions

$n \in \mathbb{Z}$ throughout.

EQUATION	GENERAL SOLUTION
$\sin \theta = 0$	$\theta = n\pi$
$\cos \theta = 0$	$\theta = (2n + 1)\frac{\pi}{2}$
$\tan \theta = 0$	$\theta = n\pi$
$\sin \theta = \sin \alpha$	$\theta = n\pi + (-1)^n \alpha$
$\cos \theta = \cos \alpha$	$\theta = 2n\pi \pm \alpha$
$\tan \theta = \tan \alpha$	$\theta = n\pi + \alpha$
$\sin^2 \theta = \sin^2 \alpha$ (also $\cos^2, \tan^2$)	$\theta = n\pi \pm \alpha$
• $\sin \theta = 1 \Rightarrow \theta = (4n + 1)\frac{\pi}{2}; \sin \theta = -1 \Rightarrow \theta = (4n - 1)\frac{\pi}{2}.$ • $\cos \theta = 1 \Rightarrow \theta = 2n\pi; \cos \theta = -1 \Rightarrow \theta = (2n + 1)\pi.$ • If $\sin \theta = \sin \alpha$ and $\cos \theta = \cos \alpha$ together, then $\theta = 2n\pi + \alpha.$	

20 Solving Standard Equation Types

- **Factorisable:** bring everything to one side and factorise. $\sin 2\theta = \sin \theta \Rightarrow \sin \theta(2 \cos \theta - 1) = 0$.
- **Quadratic in one ratio:** substitute $t = \sin \theta$ (or $\cos \theta$), solve, then **reject any root with** $|t| > 1$.
- **Same argument:** convert everything to a single ratio using $\sin^2 + \cos^2 = 1$.
- **Different arguments:** use the sum→product formulas to create a common factor.
- **Homogeneous** in $\sin \theta, \cos \theta$: divide throughout by $\cos^n \theta$ to get a polynomial in $\tan \theta$.

Two fatal errors. (1) **Never cancel** a common factor — you delete solutions. In $\sin 2\theta = \sin \theta$, cancelling $\sin \theta$ loses $\theta = n\pi$.
(2) **Squaring creates extraneous roots** — every answer must be substituted back into the original equation.

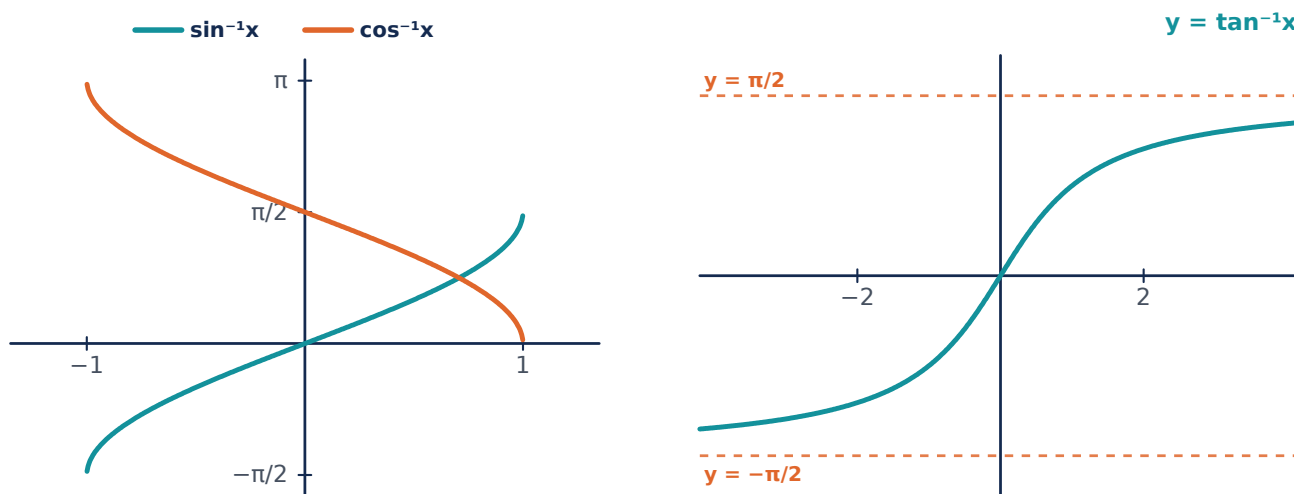
21 The Equation $a \cos \theta + b \sin \theta = c$

Divide throughout by $\sqrt{a^2 + b^2}$ and set $\cos \alpha = \frac{a}{\sqrt{a^2 + b^2}}$, $\sin \alpha = \frac{b}{\sqrt{a^2 + b^2}}$:

$$\cos(\theta - \alpha) = \frac{c}{\sqrt{a^2 + b^2}} \implies \theta = 2n\pi \pm \cos^{-1}\left(\frac{c}{\sqrt{a^2 + b^2}}\right) + \alpha$$

Solvability condition: a solution exists **if and only if** $c^2 \leq a^2 + b^2$, i.e. $|c| \leq \sqrt{a^2 + b^2}$. If $|c|$ exceeds the amplitude, the equation has **no** solution — check this *before* you start solving.

22 Inverse Functions — Principal Values



Principal-value branches. $\sin^{-1} x$ rises from $-\pi/2$ to $\pi/2$; $\cos^{-1} x$ falls from π to 0 ; $\tan^{-1} x$ has horizontal asymptotes $y = \pm\pi/2$.

FUNCTION	DOMAIN	RANGE (PRINCIPAL BRANCH)
$\sin^{-1} x$	$[-1, 1]$	$[-\frac{\pi}{2}, \frac{\pi}{2}]$
$\cos^{-1} x$	$[-1, 1]$	$[0, \pi]$
$\tan^{-1} x$	$\mathbb{R}$	$(-\frac{\pi}{2}, \frac{\pi}{2})$
$\cot^{-1} x$	$\mathbb{R}$	$(0, \pi)$
$\sec^{-1} x$	$ x \geq 1$	$[0, \pi] - \{\frac{\pi}{2}\}$
$\operatorname{cosec}^{-1} x$	$ x \geq 1$	$[-\frac{\pi}{2}, \frac{\pi}{2}] - \{0\}$

Notation trap: $\sin^{-1} x$ is **not** $\frac{1}{\sin x}$. The reciprocal is $\operatorname{cosec} x$; the -1 here denotes the inverse *function*.

23 Elementary Properties of the Inverse Functions

Cancellation — where it is legal

- $\sin(\sin^{-1} x) = x$ for every $x \in [-1, 1]$ (always true).
- $\sin^{-1}(\sin \theta) = \theta$ only if $\theta \in [-\frac{\pi}{2}, \frac{\pi}{2}]$; $\cos^{-1}(\cos \theta) = \theta$ only if $\theta \in [0, \pi]$; $\tan^{-1}(\tan \theta) = \theta$ only if $\theta \in (-\frac{\pi}{2}, \frac{\pi}{2})$.

Negative argument

- **Odd:** $\sin^{-1}(-x) = -\sin^{-1} x$, $\tan^{-1}(-x) = -\tan^{-1} x$, $\operatorname{cosec}^{-1}(-x) = -\operatorname{cosec}^{-1} x$.
- **Supplementary:** $\cos^{-1}(-x) = \pi - \cos^{-1} x$, $\cot^{-1}(-x) = \pi - \cot^{-1} x$, $\sec^{-1}(-x) = \pi - \sec^{-1} x$.

Complementary pairs

$$\sin^{-1} x + \cos^{-1} x = \frac{\pi}{2}$$

$$\tan^{-1} x + \cot^{-1} x = \frac{\pi}{2}$$

$$\sec^{-1} x + \operatorname{cosec}^{-1} x = \frac{\pi}{2}$$

Reciprocal argument

- $\sin^{-1} \frac{1}{x} = \operatorname{cosec}^{-1} x$, $\cos^{-1} \frac{1}{x} = \sec^{-1} x$ ($|x| \geq 1$).
- $\tan^{-1} \frac{1}{x} = \cot^{-1} x$ if $x > 0$, but $\tan^{-1} \frac{1}{x} = \cot^{-1} x - \pi$ if $x < 0$.

The single most-tested trap: $\sin^{-1}(\sin \theta)$ is **not** always θ . For example $\sin^{-1}(\sin 2.5) = \pi - 2.5 \approx 0.64$, not 2.5, because 2.5 lies outside $[-\frac{\pi}{2}, \frac{\pi}{2}]$. Always drag the angle back into the principal branch first.

24 Inverse Functions — Sum, Difference & Multiple Angles

Sum and difference — mind the conditions

$$\tan^{-1} x + \tan^{-1} y = \begin{cases} \tan^{-1} \frac{x+y}{1-xy}, & xy < 1 \\ \pi + \tan^{-1} \frac{x+y}{1-xy}, & x > 0, y > 0, xy > 1 \\ -\pi + \tan^{-1} \frac{x+y}{1-xy}, & x < 0, y < 0, xy > 1 \end{cases}$$

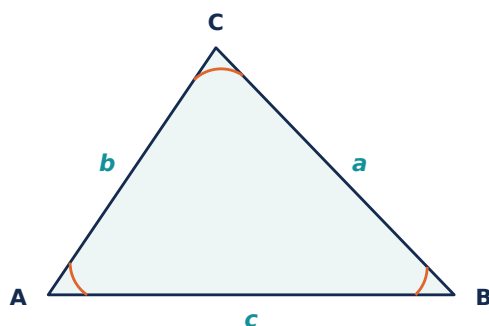
- $\tan^{-1} x - \tan^{-1} y = \tan^{-1} \frac{x-y}{1+xy}$ (valid when $xy > -1$).
- $\sin^{-1} x + \sin^{-1} y = \sin^{-1} (x\sqrt{1-y^2} + y\sqrt{1-x^2})$ (when $x^2 + y^2 \leq 1$).
- $\cos^{-1} x + \cos^{-1} y = \cos^{-1} (xy - \sqrt{1-x^2}\sqrt{1-y^2})$ (when $x + y \geq 0$).

Multiple angles — each with its own range

FORMULA	VALID WHEN
$2 \tan^{-1} x = \tan^{-1} \frac{2x}{1-x^2}$	$ x < 1$
$2 \tan^{-1} x = \sin^{-1} \frac{2x}{1+x^2}$	$ x \leq 1$
$2 \tan^{-1} x = \cos^{-1} \frac{1-x^2}{1+x^2}$	$x \geq 0$
$2 \sin^{-1} x = \sin^{-1} (2x\sqrt{1-x^2})$	$ x \leq \frac{1}{\sqrt{2}}$
$2 \cos^{-1} x = \cos^{-1} (2x^2 - 1)$	$x \geq 0$
$3 \sin^{-1} x = \sin^{-1} (3x - 4x^3)$	$ x \leq \frac{1}{2}$
$3 \cos^{-1} x = \cos^{-1} (4x^3 - 3x)$	$\frac{1}{2} \leq x \leq 1$
$3 \tan^{-1} x = \tan^{-1} \frac{3x-x^3}{1-3x^2}$	$ x < \frac{1}{\sqrt{3}}$

The condition is part of the formula. Outside its range each identity needs a $\pm\pi$ correction — examiners choose the values deliberately, so check the range *before* you apply the result.

25 Properties of Triangles



Standard notation — side a is opposite angle A , b opposite B , c opposite C . Every formula below uses this convention.

Sine rule

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} = 2R$$

Cosine rule

$$\cos A = \frac{b^2 + c^2 - a^2}{2bc}, \quad a^2 = b^2 + c^2 - 2bc \cos A$$

Projection rule

$$a = b \cos C + c \cos B$$

$$b = c \cos A + a \cos C$$

$$c = a \cos B + b \cos A$$

Napier's analogy

$$\tan \frac{A - B}{2} = \frac{a - b}{a + b} \cot \frac{C}{2}$$

26 Area, Half-Angles & Associated Circles

With $s = \frac{a + b + c}{2}$ (the semi-perimeter) and Δ the area:

$$\Delta = \frac{1}{2}ab \sin C = \sqrt{s(s-a)(s-b)(s-c)} = \frac{abc}{4R} = rs$$

Half-angle formulae

$$\sin \frac{A}{2} = \sqrt{\frac{(s-b)(s-c)}{bc}}$$

$$\cos \frac{A}{2} = \sqrt{\frac{s(s-a)}{bc}}$$

$$\tan \frac{A}{2} = \sqrt{\frac{(s-b)(s-c)}{s(s-a)}}$$

Circumradius, inradius, exradii

- $R = \frac{abc}{4\Delta} = \frac{a}{2 \sin A}$
- $r = \frac{\Delta}{s} = (s-a) \tan \frac{A}{2} = 4R \sin \frac{A}{2} \sin \frac{B}{2} \sin \frac{C}{2}$
- $r_1 = \frac{\Delta}{s-a} = s \tan \frac{A}{2} = 4R \sin \frac{A}{2} \cos \frac{B}{2} \cos \frac{C}{2}$ (similarly r_2, r_3)
- $r_1 + r_2 + r_3 - r = 4R$; $\frac{1}{r} = \frac{1}{r_1} + \frac{1}{r_2} + \frac{1}{r_3}$; $\Delta = \sqrt{r r_1 r_2 r_3}$

27 Quick Recall & The Traps That Cost Marks

- **Radians only** for $s = r\theta$, sector area, limits and derivatives.
- $\tan$ and $\cot$ have period π ; the other four have 2π . $|\sin \theta|$ has period π .
- Allied angles: name changes only for 90° and 270° ; the sign comes from ASTC.
- In $\cos C - \cos D$ there is a **leading minus sign**.
- The $\pm$ in a half-angle formula is fixed by the quadrant of $\frac{A}{2}$, not of A .
- $\sin \theta = \sin \alpha \Rightarrow \theta = n\pi + (-1)^n \alpha$ — **not** $2n\pi \pm \alpha$ (that one is for cosine).
- **Never cancel** a factor in a trigonometric equation; factorise instead.
- **Always verify** after squaring — squaring manufactures extraneous roots.
- $a \cos \theta + b \sin \theta = c$ is solvable **only if** $c^2 \leq a^2 + b^2$.
- Reject any root that makes $\tan$, $\sec$, $\cot$ or cosec undefined, or forces $|\sin|, |\cos| > 1$.
- **Inverse:** $\sin^{-1} x \neq \frac{1}{\sin x}$; and $\sin^{-1}(\sin \theta) = \theta$ **only** inside the principal branch.
- **Inverse:** every sum / multiple-angle formula for $\tan^{-1}$ and $\sin^{-1}$ carries a **range condition** — check it before using the formula.
- $\sin^{-1} x + \cos^{-1} x = \frac{\pi}{2}$ holds only for $x \in [-1, 1]$; $\tan^{-1} x + \cot^{-1} x = \frac{\pi}{2}$ holds for all real x .